

 Handling Outliers & Z-Scores Covariance & Correlation

INSTANT

 Data Analysis Diploma • Session 04

DESCRIPTIVE STATISTICS

Handling Outliers · Z-Score · Covariance & Correlation · Statistics & Date/Time
Functions in Excel

6 Goals**Core Outcomes**

Outliers • Z-Score • Excel

52 Slides**Curriculum Depth**

Part 1 & Part 2 Unified

Visual Labs**Hands-on Practice**

Gaussian Curve & Worksheets

Start Session 04 → Excel Statistical Functions Date & Time Arithmetic

— PART 1 COMPETENCIES

What You Will Be Able To Do

By the end of this session, you will confidently apply these analytical capabilities to real datasets:



01

Identify Outliers

Recognize abnormal values in a dataset using both visual and mathematical methods



02

Understand Their Impact

See how outliers distort mean, variance, and downstream business analysis



03

Calculate Z-Scores

Measure how far a value sits from the mean on a standardized standard-deviation scale



04

Read Relationships

See how two variables move together using bivariate scatter plots and trend vectors



05

Covariance vs Correlation

Know the critical difference in units, scale, comparability, and when to report each



06

Support Decisions

Translate statistical relationship metrics into actionable, executive-ready business insights



PART 1

Quick Recap

Part 1 in sixty seconds — the vocabulary today's session builds on.

— CONCEPTS IN 60 SECONDS

From Descriptive Statistics Part 1

Essential statistical vocabulary and measures from Session 03 that form the bedrock for today:



Mean

The balancing point of all values: $\Sigma x \div n$



Median

The middle value in sorted order
(resistant to extremes)



Mode

The most frequently occurring
observation in data



Variance & SD

How spread out values are from the
mean (s^2 & s)



Quartiles

Q1, Median, Q3 — dividing the sorted
distribution into fourths



Percentiles

The exact value below which a given % of
data falls



Box Plot

Five-number summary visually capturing
center and spread

💡 These foundations enable us to quantify distributions and rigorously flag deviations from the expected norm.



PART 2

Handling Outliers

One unusual value can quietly reshape an entire analysis.

DEFINITION & EXAMPLES

What Are Outliers?

A data point significantly different from other observations in the distribution.



DEFINITION & PROPERTIES

A data point significantly different from other observations in the distribution.

- Sits far outside the typical pattern of the rest of the dataset
- Can be a genuine extreme real-world case, or a data entry/sensor error worth checking
- Not automatically "wrong" — always investigate root causes before deciding to remove or retain
- Every dataset should be systematically scanned for outliers before statistical modeling begins

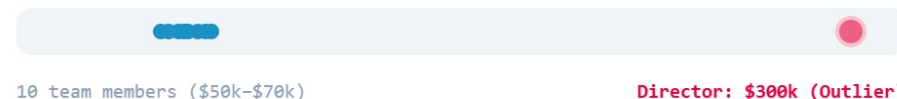
Crucial Rule: Never delete an outlier without verifying root cause (entry typo vs real black-swan event).



Salary Example

HR & PAYROLL COMPENSATION

A team of 10 engineers earning \$50k–\$70k, with one executive director earning \$300k



Sales Example

E-COMMERCE REVENUE SPIKE

Typical daily e-commerce sales hovering near \$2,000, with one Black Friday spike to \$40,000



— CLASSIFICATION

Three Types of Outliers

Understanding the contextual nature of anomalies to determine proper treatment:



TYPE 01

Global Outlier

Point Anomaly

- Extreme compared to the entire dataset regardless of circumstances
- Example: one \$300k salary among an entry-level team earning \$50k–\$70k
- The most common and easiest type of outlier to spot on basic charts

🔍 Easiest to detect using Box Plot & Z-Scores.



TYPE 02

Contextual Outlier

Conditional Anomaly

- Unusual only within a specific temporal, seasonal, or behavioral context
- Example: 35°C is normal in summer, but a severe anomaly in winter
- Requires knowing domain context and segmenting data to detect

📅 Requires conditional grouping (e.g. Month/Country).



TYPE 03

Collective Outlier

Pattern Anomaly

- A group of values that together deviate noticeably from normal structure
- Example: a sudden cluster of failed transactions within a 15-minute window
- Individually each value may look valid, but the collective pattern is abnormal

🕒 Detected via time-series sequence models.

— DISTORTION RISKS

Why Outliers Matter

How extreme values propagate through pipelines, skewing conclusions and business decisions:



The Mean

Pulled sharply toward the outlier, misrepresenting what is typical for the majority

Risk Factor #1



Standard Deviation

Inflates rapidly, making normal baseline data look far more volatile than it is

Risk Factor #2



ML Models

Distorts trained regression gradients and reduces predictive precision on normal cases

Risk Factor #3



Business Decisions

Wrong analytical conclusions can lead to costly, misdirected executive budget allocations

Risk Factor #4



Without outlier: **Mean = \$60k** | With single \$300k outlier: **Mean jumps to \$84k (+40% false inflation!)**

Median remains steady at \$60k (Resistant)

— 4 SCREENING METHODS

How Do We Detect Outliers?

Four industry-standard methods from exploratory visualization to rigorous parametric tests:



METHOD 01

Visualization

Scatter plots and histograms make isolated extreme points immediately visible at a glance



METHOD 02

Box Plot

Points falling beyond the $1.5 \times \text{IQR}$ whiskers are mathematically flagged automatically



METHOD 03

IQR Rule

A precise, non-parametric formula calculating lower and upper fence boundaries



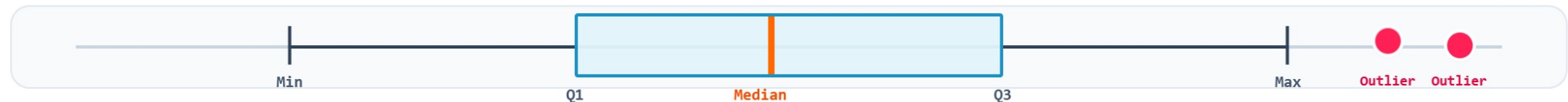
METHOD 04

Z-Score

Parametric standard deviation threshold flagging any standardized score with $|Z| > 3$

Visual Box Plot Whisker Rule

WHISKER LIMIT = $1.5 \times \text{IQR}$



— INNER & OUTER FENCES

The IQR Rule

Any value beyond the fences is flagged as a potential outlier — here, 45 clearly falls outside the upper fence.

MATHEMATICAL CRITERION

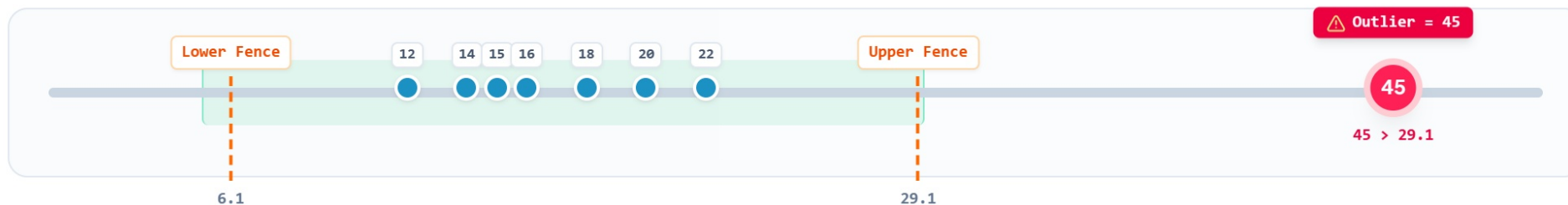
Outlier if value $< Q1 - 1.5 \times IQR$ or value $> Q3 + 1.5 \times IQR$

Lower Fence: 6.1

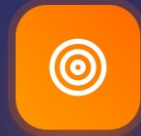
Upper Fence: 29.1

Calibrated Boundary Number Line

Observation 45 flagged outside upper fence



Any value falling beyond the inner fences is flagged for review — here, 45 sits far outside the upper fence of 29.1.



PART 3

Z-Score

"How many standard deviations away from the mean does this value sit?"

— EMPIRICAL RULE (68-95-99.7)

Z-Score — Standardizing a Value

Converts any raw observation into a dimensionless, universally comparable unit of spread.

DEFINITION & FORMULA

Formula:

$$Z = (X - \text{Mean}) \div \text{Standard Deviation}$$

$$Z = (X - \mu) \div \sigma$$

- Measures how many standard deviations a value is located away from the mean
- Formula: $Z = (X - \text{Mean}) \div \text{Standard Deviation}$
- Converts any value into a common, comparable scale across different datasets and units
- Works especially well when data follows a roughly normal (bell-shaped) distribution
- The fundamental building block for statistically detecting abnormal outliers

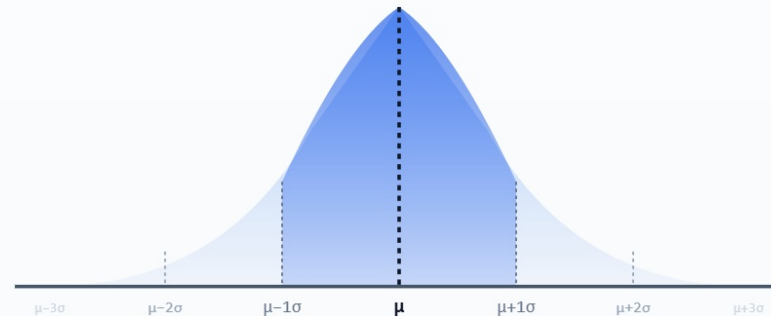
💡 A Z-score of +2.0 means the value sits exactly 2

Empirical Rule (68 – 95 – 99.7)

Normal distribution standard deviation intervals

BELL CURVE

68% within $\pm 1\sigma$ • 95% within $\pm 2\sigma$ • 99.7% within $\pm 3\sigma$



$\pm 1\sigma$ Spread

$\pm 2\sigma$ Spread

$\pm 3\sigma$ Spread

— POLARITY & MEAN RELATIVE POSITION

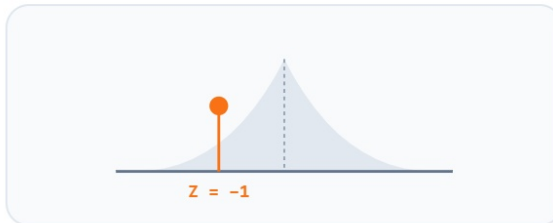
Positive, Negative & Zero Z-Scores

A negative Z-score means the value is below the mean; positive means above; zero means it equals the mean exactly.

BELOW THE MEAN

Negative Z-Score

Value is smaller than average

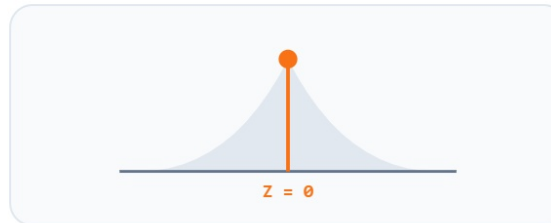


Value sits to the left of center, below average.

EXACTLY AT CENTER

Zero Z-Score

Value equals the mean

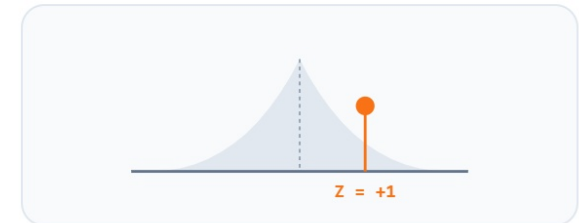


Value is perfectly typical, coinciding with μ .

ABOVE THE MEAN

Positive Z-Score

Value is greater than average



Value sits to the right of center, above average.

A negative Z-score means the value is below the mean; positive means above; zero means it equals the mean exactly.

— THRESHOLD CLASSIFICATION

Interpreting Z-Score Values

Translating numeric Z-scores into statistical frequency and outlier likelihood:

Standard Interpretation Matrix

Mapping standardized scores to probabilistic rarity

GAUSSIAN THRESHOLDS

Z-Score	Meaning	Interpretation
$Z = 0$	The value equals the mean exactly	Perfectly typical — right at the center of the distribution
$Z = +1$	One standard deviation above the mean	Somewhat above average, still very common (~84th percentile)
$Z = -1$	One standard deviation below the mean	Somewhat below average, still very common (~16th percentile)
$ Z > 3$	More than three standard deviations from the mean	Rare (<0.3% probability) — commonly flagged as an outlier

● Tail Flag Example: Observation at $Z = +3.5$ sits in the top 0.02% of data — flagged as extreme outlier.

 $|Z| > 3$ Rule Triggered

— STEP-BY-STEP CALCULATION

Step-by-Step: Calculating a Z-Score

Dataset: 10, 12, 15, 18, 20 — Calculate the Z-score for the value 20

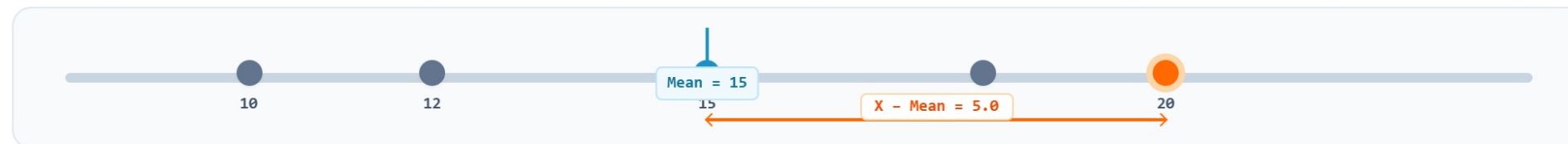
WORKED EXAMPLE

Dataset: 10, 12, 15, 18, 20 — Calculate the Z-score for the value 20

$X = 20$

$\mu = 15$

$\sigma \approx 3.69$



STEP 1: DISTANCE

$$X - \text{Mean} = 20 - 15 = 5.0$$

How many raw units above center

STEP 2: DIVIDE BY SD

$$5.0 \div 3.69 \approx 1.35$$

Normalize into SD yardsticks

STEP 3: RESULT & MEANING

$$Z = +1.35 \text{ (Not an outlier)}$$

Well inside standard $|Z| < 3$ zone

— THRESHOLD RULE ($|Z| > 3$)

Using Z-Score to Flag Outliers

Deploying standardized thresholds for automated anomaly detection pipelines:

DETECTION IN PRACTICE

Threshold Rule:

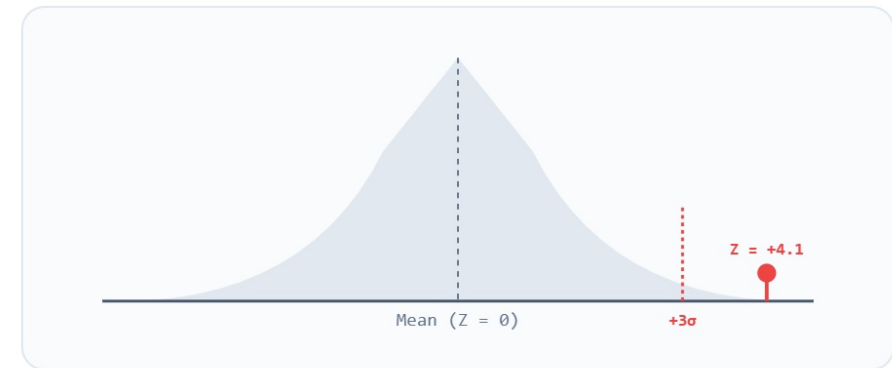
Flag as outlier if $|Z| > 3.0$

- Calculate the Z-score for every single numeric value in the target column
- Common rule of thumb: flag any value with $|Z| > 3$ as a potential outlier
- Works best and most reliably when data follows a roughly normal distribution
- Business example: an automated transaction 4 standard deviations above user average triggers fraud hold
- Provides a precise, repeatable, code-friendly alternative to manual visual inspection

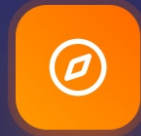
🛡️ Used extensively in fintech fraud detection: unusual transaction amounts trigger automatic security holds.

Visualizing the 3σ Cutoff Zone

Any point past $\pm 3\sigma$ represents $<0.3\%$ baseline probability



🚩 **Z = +4.1 Flagged:** Sits 4.1 standard deviations above normal — virtually impossible under normal random chance.



PART 4

Covariance

Do two variables tend to rise and fall together — or in opposite directions?

— DIRECTION ONLY

Covariance — Do Two Variables Move Together?

Quantifying whether deviations from respective means align in unison or oppose each other.

COVARIANCE DEFINITION

Measures the directional alignment of joint variability between two variables.

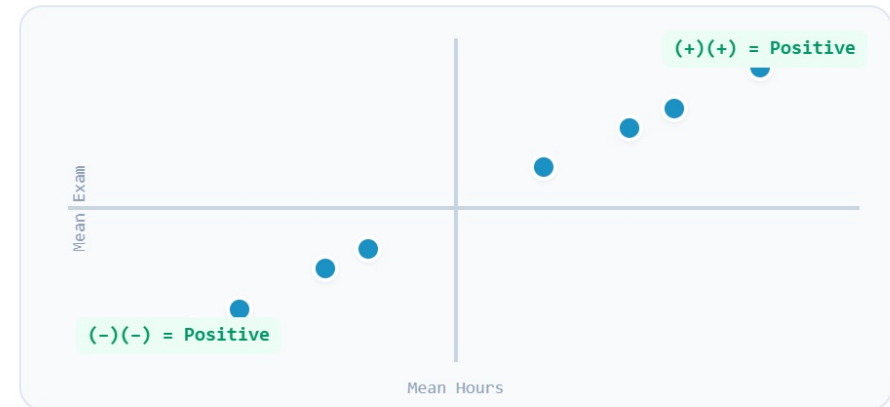
- Positive covariance: variables tend to increase together (both above mean or both below)
- Negative covariance: as one variable increases, the other tends to decrease
- Zero covariance: no consistent relationship in directional movement
- Crucial limitation: only indicates direction — not how strong the relationship actually is
- Excel: =COVARIANCE.P() for a complete population
- Excel: =COVARIANCE.S() for an empirical sample

Population: =COVARIANCE.P()

Sample: =COVARIANCE.S()

Quadrant Analysis: Study Hours vs Exam Score

BIVARIATE



When values exceed their respective means together, their product is positive, yielding positive covariance.

— POSITIVE CO-MOVEMENT

Positive Covariance

When above-average values in X correspond to above-average values in Y.

DIRECT RELATIONSHIP

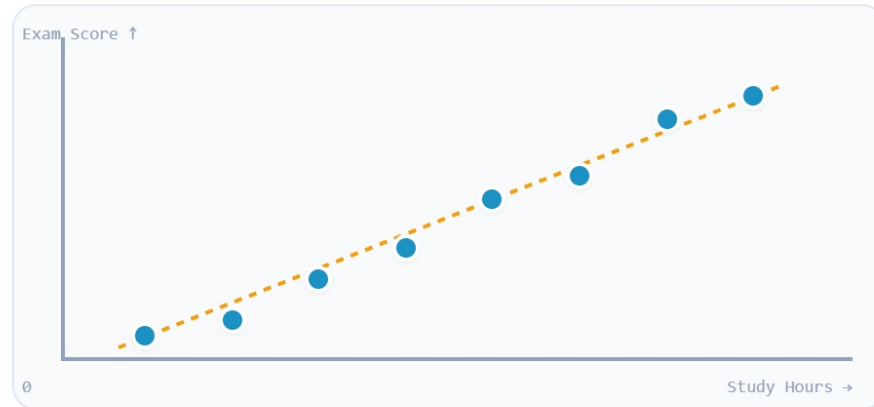
Study Hours vs Exam Scores

- As study hours increase, exam scores tend to systematically increase too
- Both variables rise together — data points cluster in the "both high" and "both low" quadrants
- Business example: Marketing Spend vs Revenue — higher monthly ad budget drives higher sales
- Business example: Employee Training Hours vs Operational Quality Score

 **Directional Rule:** When X goes UP (↑), Y also tends to go UP (↑).

Positive Trend Scatter Plot

UPWARD SLOPE



Points cluster along an upward vector, indicating positive covariance.

INVERSE CO-MOVEMENT

Negative Covariance

When an increase in one variable corresponds systematically to a decrease in the other.

INVERSE RELATIONSHIP

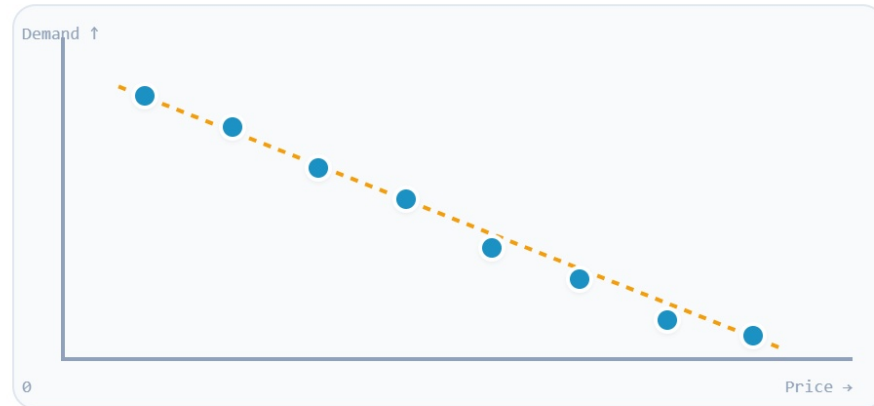
Price vs Demand

- As product price increases, market consumer demand tends to decrease
- One variable rises while the other falls — the classic downward economic relationship
- Business example: Product retail price vs units sold per month
- Business example: Shipping delivery delay hours vs customer CSAT satisfaction score

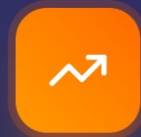
 **Directional Rule:** When X goes UP (↑), Y tends to go DOWN (↓).

Negative Trend Scatter Plot

DOWNWARD SLOPE



As product price escalates, consumer demand contracts systematically.



PART 5

Correlation

The strength and direction of a relationship, on one universal scale.

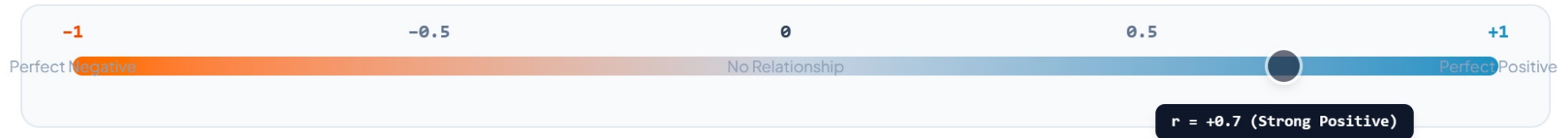
— NORMALIZED FROM -1 TO +1

Correlation — Strength and Direction

Correlation measures both the strength and direction of the relationship between two variables, always bounded on the same -1 to +1 scale — regardless of original units.

The Universal Standard: Pearson Correlation (r)

Correlation measures both strength and direction, always strictly normalized between -1.0 and +1.0 regardless of original units.



+1.0
Perfect Positive
 Points fall exactly on an upward straight line

0.0
No Linear Relationship
 Variables move completely independently

-1.0
Perfect Negative
 Points fall exactly on a downward straight line

COHEN & INDUSTRY STANDARDS

Interpreting Correlation Strength

Industry-standard guidelines to interpret Pearson correlation coefficient (r):

Benchmark Interpretation Guide

Standard rule-of-thumb tiers for business analysis

PEARSON CLASSIFICATION

Correlation Range	Strength	Business Example
0.8 to 1.0 (or -0.8 to -1.0)	Very Strong	Digital advertising spend and impressions delivered
0.6 to 0.8 (or -0.6 to -0.8)	Strong	Paid marketing spend and monthly sales revenue
0.4 to 0.6 (or -0.4 to -0.6)	Moderate	Employee job tenure and annual performance rating
0.2 to 0.4 (or -0.2 to -0.4)	Weak	Office ambient temperature and programmer productivity
0.0 to 0.2 (or -0.0 to -0.2)	Very Weak / None	Employee shoe size and annual base salary

⚠ Remember: Correlation implies association, not causation. Always investigate business context before drawing causal conclusions.

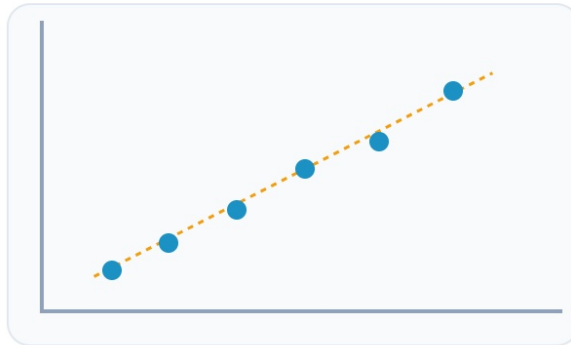
— THREE CANONICAL ARCHETYPES

Reading a Scatter Plot

Visually identifying direction and tightness of clustering around linear trends:

POSITIVE CORRELATION

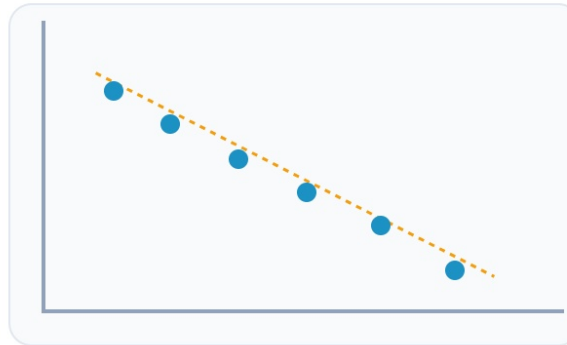
Sales vs Marketing Spend



Upward diagonal alignment: high spend reliably pairs with high revenue.

NEGATIVE CORRELATION

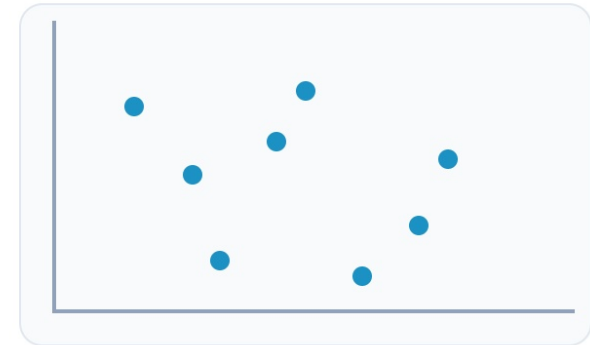
Demand vs Price



Downward diagonal alignment: raising prices decreases total market volume.

NO CORRELATION ($r \approx 0$)

Salary vs Shoe Size



Scattered cloud without discernible slope; linear correlation coefficient is ~ 0 .

EXECUTIVE REFERENCE MATRIX

Covariance vs Correlation

Clear dimensional differences between raw unstandardized co-movement and standardized correlation:

Executive Comparison Matrix

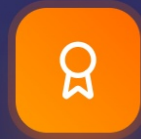
Choosing the appropriate metric for technical vs reporting workflows

SIDE-BY-SIDE

Criteria	Covariance	Correlation (r)
What it shows	Direction only (+, −, or 0)	Direction AND standardized strength
Range	No fixed limits — depends entirely on original units of X and Y	Strictly bounded between −1.0 and +1.0
Comparability	Hard to compare across different variable pairs (e.g. \$ vs hours)	Universally comparable across any metric pairs in the business
Typical use	Initial mathematical intermediate step in portfolio & linear algebra math	The primary metric reported on executive dashboards and reports

Key Formula Link:

$$r = \text{Cov}(X, Y) \div (\text{SD}(X) \times \text{SD}(Y))$$



PART 6

Hands–On Activity

Marketing Spend vs Sales — calculate, interpret, and discuss as a group.

8 MONTH OBSERVATIONS

Dataset — Marketing Spend vs Sales

Monthly performance data tracking advertising investment and resulting revenue:

Monthly Advertising & Sales Observations

Dataset tracked across 8 consecutive operating months (in thousands)

n = 8 OBSERVATIONS

Month	Marketing Spend (\$'000)	Sales (\$'000)
January	\$10k	\$105k
February	\$15k	\$120k
March	\$12k	\$110k
April	\$20k	\$140k
May	\$25k	\$160k
June	\$18k	\$130k
July	\$22k	\$150k

— 6 ANALYTICAL STEPS

Your Turn — Calculate & Interpret

Work through these six sequential steps to generate strategic business recommendations:

1

Solved ✓

Calculate Means
 Calculate the mean Marketing Spend and mean Sales across the 8 months

 Mean Spend = 19.0 (\$19k), Mean Sales = 136.25 (\$136.25k)

2

Click to solve

Calculate Covariance
 Calculate the sample covariance between Marketing Spend and Sales

 Click to reveal calculation

3

Click to solve

Calculate Correlation (r)
 Calculate the Pearson correlation coefficient between them

 Click to reveal calculation

4

Click to solve

Visualize Trend
 Sketch or verify what the scatter plot would likely look like

 Click to reveal calculation

5

Click to solve

Classify Strength
 Classify the strength: weak, moderate, strong, or very strong?

 Click to reveal calculation

6

Click to solve

Executive Recommendation
 Write a one-sentence business recommendation based on your result

 Click to reveal calculation

Active Step: 1 of 6

Reveal All Solutions →

— PART 1 RECAP

What We Covered in Part 1

Core principles of anomaly detection and relationship analysis mastered in this part:



01

Outliers

Global, contextual & collective anomalies — and why they distort models and means



02

IQR Rule

$Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$ non-parametric fences to flag rogue points



03

Z-Score

Standardized deviations from the mean; flags extreme points where $|Z| > 3$



04

Covariance

Measures directional co-movement between two variables (+, -, or 0)



05

Correlation

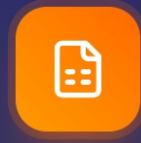
Measures both direction AND standardized strength on a universal -1 to +1 scale



06

Applied It

Real-world Marketing Spend vs Sales calculated, visualized, and translated into strategy



PART 2

Statistics & Date/Time Functions in Excel

AVERAGE · STDEV.S · PERCENTILE · RANK · DATEDIF · NETWORKDAYS · and
more • Core Excel formulas for real analysis

— EXCEL COMPETENCIES

What You Will Be Able To Do

Core applied capabilities you will master with Excel functions in this part:



01

Key Statistical Functions

Know the most important Excel statistics functions for daily reporting and aggregations



02

How They Work

Understand the syntax, arguments, and underlying logic behind each formula



03

Date & Time Functions

Extract, build, arithmetic-shift, and compare calendar dates and time stamps



04

Business Use Cases

Apply both function families directly to solve real-world executive business questions



MODULE 1

Introduction

Why Excel's statistical functions are essential to every data analyst's toolkit.

— FROM DATA TO DECISIONS

Why Excel Statistical Functions Matter

Formulas turn raw spreadsheet columns into automated, decision-ready answers.

VALUE PROPOSITION

Formulas turn raw spreadsheet columns into automated, decision-ready answers.

- Excel is the most widely used analysis tool in business — statistical functions are how raw rows become insight
- They calculate instantly and recalculate automatically as new data arrives
- The same functions covered in Descriptive Statistics now become tools you use every day
- Mastering a core set of ~15 functions covers the large majority of everyday analysis needs

Excel functions provide deterministic, dynamic recalculation that scales across thousands of rows.

The 3-Step Analytical Transformation

PIPELINE



1. Raw Spreadsheet Data

Unprocessed rows and columns exported from systems



2. Apply a Statistical Function

Syntax-driven automated aggregation formula



3. Decision-Ready Business Answer

Instant clarity ready for executive reports & dashboards

— 5 BUSINESS VERTICALS

Real-World Applications Across the Business

How statistical formulas power daily operations and reporting across departmental verticals:



Sales Analysis

Average deal size, top performers, monthly variation

Sector 01



HR Analytics

Salary bands, tenure distribution, headcount trends

Sector 02



Finance

Budget variance, expense ranking, quarter-end reporting

Sector 03



Marketing

Campaign performance, conversion percentiles

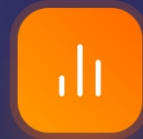
Sector 04



Operations

Delivery times, working-day scheduling, capacity limits

Sector 05



MODULE 2

Statistical Functions

Fifteen functions that turn a column of numbers into a business answer.

— AVERAGE • MEDIAN • MODE

Central Tendency: AVERAGE • MEDIAN • MODE.SNGL

Primary formulas to identify the balancing point, midpoint, and most frequent observation:

CENTRAL TENDENCY FUNCTIONS

EXCEL SYNTAX

AVERAGE()

MEDIAN()

MODE.SNGL()

Selected Syntax:

=AVERAGE(B2:B9)

Calculated Result:

44.0

AVERAGE()

44.0

Typical daily units sold across the sample week

MEDIAN()

44.5

A more robust "typical" figure when a big order skews the average

MODE.SNGL()

47

The single most common order size — useful for stock planning

Workbook Simulation

Sheet1.xlsx

fx

=AVERAGE(B2:B9)

	A	B
1		Units Sold
2		42
3		38
4		55
5		61
6		29
7		47
8		33
9		47
10	Result	44.0

Select any function tab above to see how Excel recalculates the dynamic range result.

— COUNT · COUNTA · COUNTBLANK

Counting: COUNT · COUNTA · COUNTBLANK

Distinguishing between numeric values, populated cells, and missing records:

DATA AUDITING FUNCTIONS

COUNT()

COUNTA()

COUNTBLANK()

Active Function:

=COUNT(B2:B10)

Total Count:

3

Counts numbers only (120, 85, 95)

⚠ **Common Interview Trap:** =COUNT() ignores text entries completely. Use =COUNTA() when counting names, IDs, or status codes.

Order Log Sheet

OrderLog.xlsx

fx

=COUNT(B2:B10)

	B (Order Log)	Type
2	Shipped	Text
3	120	Number
4	Pending	Text
5	85	Number
6	(blank)	Empty
7	Shipped	Text
8	95	Number
9	(blank)	Empty
10	Delivered	Text
11	COUNT Result:	3

— MAX · MIN · STDEV.S · VAR.S

Range & Spread: MAX · MIN · STDEV.S · VAR.S

Quantifying extremes, consistency, and dispersion across operational columns:

SPREAD & DISPERSION FORMULAS

MAX() 61
 =MAX(B2:B9)
 The single best-selling day in the sample

MIN() 29
 =MIN(B2:B9)
 The slowest day — worth investigating why

STDEV.S() 10.78
 =STDEV.S(B2:B9)
 How much daily sales typically swing from the average

VAR.S() 116.29
 =VAR.S(B2:B9)
 The squared-unit basis STDEV.S is calculated from

Operational Spread Interpretation

How much daily volume swings from normal expectation

BEST DAY (MAX)

61

Top peak demand

SLOWEST DAY (MIN)

29

Trough volume

STANDARD DEVIATION (STDEV.S)

±10.78

Daily sales typically fluctuate ±10.78 units around the 44.0 unit mean.

$\text{VAR.S}() = (10.78)^2 =$ **116.29 squared units**

Ranking & Percentiles

Benchmarking performance, cutoffs, and ordered positional ranks:

Ranking & Percentile Formulas

Benchmarking position, cutoffs, and ordered ranks

ORDER METRICS

Function	Syntax & Example	Result	Business Use Case
PERCENTILE()	=PERCENTILE(B2:B9, 0.9)	73	The value below which 90% of results fall
QUARTILE()	=QUARTILE(B2:B9, 3)	62.5	Q3 — the top quarter of results starts here
RANK()	=RANK(40, B2:B9, 0)	5	This value's position, highest to lowest
LARGE()	=LARGE(B2:B9, 2)	70	The second-highest value in the range
SMALL()	=SMALL(B2:B9, 2)	20	The second-lowest value in the range

💡 **LARGE vs SMALL:** =LARGE(range, 2) fetches second-highest; =SMALL(range, 2) fetches second-lowest.



MODULE 3

Date Functions

Every business report runs on dates — orders, deadlines, and reporting periods.

— TODAY · NOW · DATE

Current Date & Building Dates

Generating volatile system timestamps and assembling clean dates from fragmented columns:

DATE GENERATION FUNCTIONS

TODAY() 26-Jan-2026

=TODAY()

NOW() 26-Jan-2026 09:41 AM

=NOW()

DATE() 14-Mar-2026

=DATE(2026, 3, 14)

Business Scenario: A live dashboard opens with =TODAY() to show "days since order" without manual updates. =DATE() rebuilds a proper date when year, month and day arrive in separate columns — common with system exports.

Live Dynamic Calendar (January 2026)

TODAY() = 26-Jan-2026

Sun	Mon	Tue	Wed	Thu	Fri	Sat
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

NOW() Live Clock: 26-Jan-2026 09:41 AM

— DAY • MONTH • YEAR

Extracting Date Components

Deconstructing datetime timestamps into analytical cohorts (Year, Month, Day):

DATE DRILLDOWN

DAY() 14
=DAY(A2)

MONTH() 3
=MONTH(A2)

YEAR() 2026
=YEAR(A2)

Business Scenario: Extracting MONTH() and YEAR() from every order date is the first step in building any monthly or yearly sales trend report.

Deconstructing Date: 14-Mar-2026

YEAR 2026

MONTH 3 (March)

DAY 14

14-Mar-2026 is built from three nested components — broadest to most specific.

— EDATE · EOMONTH

Date Math: EDATE() · EOMONTH()

Calculating forward/backward milestones and month-end financial closing dates:

Date Math: EDATE() & EOMONTH()

Projecting contract milestones and accounting period cutoffs



=EDATE(start_date, months)

Jumps forward or backward by exact month intervals while preserving the same day of the month.

=EOMONTH(start_date, months)

Returns the exact last day of the month, accounting automatically for 28, 30, 31, and leap years.



MODULE 4

Time Functions

Timestamps carry as much business information as dates do.

— TIME · HOUR · MINUTE · SECOND

TIME() · HOUR() · MINUTE() · SECOND()

Isolating operational hours, customer wait times, and building clean intraday timestamps:

TIME EXTRACTION & ASSEMBLY

TIME()

=TIME(14, 35, 20)

2:35:20 PM

HOUR()

=HOUR(A2)

14

MINUTE()

=MINUTE(A2)

35

SECOND()

=SECOND(A2)

20

Business Example: A call center logs ticket timestamps and uses HOUR() to identify peak call times, while TIME() assembles a clean timestamp from separately captured hour, minute and second fields.

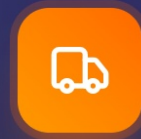
Timestamp: 14:35:20 (2:35:20 PM)



HOUR: 14

MIN: 35

SEC: 20



MODULE 5

Date Difference Functions

Turning two dates into a duration — the most business-critical date skill.

— DATEDIF • NETWORKDAYS • WORKDAY

DATEDIF() • NETWORKDAYS() • WORKDAY()

Measuring elapsed calendar days, working business days, and computing exact target deadlines:

=NETWORKDAYS(A2, B2)

A2: Start Date of Range

B2: End Date of Range

Function	Syntax & Example	Result
DATEDIF()	=DATEDIF(A2, B2, "d")	72 days between 1-Jan-2026 and 14-Mar-2026
NETWORKDAYS()	=NETWORKDAYS(A2, B2)	22 working days in January 2026
WORKDAY()	=WORKDAY(A2, 10)	15-Jan-2026 – 10 working days after 1-Jan-2026

🚩 **Difference:** =DATEDIF() calculates raw elapsed days/months/years, whereas =NETWORKDAYS() automatically removes weekends and holidays.

— 4 OPERATIONAL SCENARIOS

Business Scenarios

Real-world operations formulas applied directly to customer fulfillment, HR tenure, and project schedules:



Delivery Time

=B2-A2 or NETWORKDAYS()

Measure order-to-ship duration excluding weekends



Employee Experience

=DATEDIF(HireDate, TODAY(), "y")

Calculates exact completed years of tenure



Project Duration

=NETWORKDAYS(Start, End)

Calculates real working-day effort required



Working Days Calculation

=WORKDAY(Start, N)

Projects realistic deadline dates factoring weekends

MASTER FUNCTION DIRECTORY

Quick Reference — All Functions at a Glance

Every statistical, date, and time function organized by analytical purpose:

Master Directory — All Functions at a Glance

Every formula covered in Session 04 indexed by analytical objective

22 FUNCTIONS

Category	Functions
Center & Frequency	AVERAGE() • MEDIAN() • MODE.SNGL()
Counting	COUNT() • COUNTA() • COUNTBLANK()
Range & Spread	MAX() • MIN() • STDEV.S() • VAR.S()
Ranking & Percentiles	PERCENTILE() • QUARTILE() • RANK() • LARGE() • SMALL()
Date Functions	TODAY() • NOW() • DATE() • DAY() • MONTH() • YEAR() • EDATE() • EOMONTH()
Time Functions	TIME() • HOUR() • MINUTE() • SECOND()
Date Difference	DATEDIF() • NETWORKDAYS() • WORKDAY()

— 4 CORE PILLARS

Session Summary

The four fundamental dimensions connecting statistical theory with practical Excel execution:



Why It Matters

Excel functions as the everyday face of statistics across all industries



Statistical Functions

15 functions covering center, spread, counting & ranking in sheets



Date & Time Functions

Building, extracting and comparing dates, timestamps, and hours



Date Differences

DATEDIF, NETWORKDAYS and WORKDAY for real operational durations

— 5 CRITICAL RULES

Key Takeaways

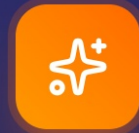
Five golden rules every data analyst must remember when applying functions in production:



Five Golden Rules for Production Analytics

Core principles every data professional must carry into practice

- 01 The right function depends on the question — MEAN for typical value, MEDIAN when outliers are present, MODE for the most common category
- 02 COUNT, COUNTA, and COUNTBLANK each answer a different data-quality question — know which one you actually need
- 03 STDEV.S and VAR.S reveal consistency, not just typical value — always use the .S versions for real-world sample data
- 04 TODAY() and NOW() are live and recalculating — never rely on them for a fixed historical record
- 05 NETWORKDAYS() and WORKDAY() are the two most business-critical date functions for deadlines and delivery promises



From Formula to Insight.

Every function you learned today turns a spreadsheet column into a decision someone can act on.

Next session: Advanced Analytical Modeling & Data Wrangling

Instant Academy • Data Analysis Training Program